Alexandre Pinto

Software Engineer
Portugal

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Summary

Senior software engineer specialized in designing and building scalable backend and distributed systems with strong proficiency across the Python ecosystem. Values system design before implementation, favouring maintainable and well-tested code.
Focused on building AI/LLM-backed product features, from concept through to production.

Work Experience

Jan 2022 - Present

Senior Backend Engineer - TestGorilla
At TestGorilla I develop and expand core product features of the platform, designing and writing efficient code that follows best industry practices regarding performance, scalability and testing (app.testgorilla.com).
My daily work consists of leading backend initiatives end-to-end, from scoping and implementation to code reviews, testing, technical documentation, and production release management.
Projects span natural-language search powered by LLM-based information extraction systems, data modelling, integration of third-party services, and the design of scalable REST APIs and backend services. This work includes core contributions to customer-facing products such as TestGorilla Sourcing (natural-language talent search) and TestGorilla Profiles (skills-based candidate profiles).
Technologies: Python 3.x, Django, Django Rest Framework, Celery, Redis, MySQL, Algolia, Milvus, LLM APIs, LangChain, Docker, CircleCI, AWS, Git.

Nov 2017 - Nov 2021

Senior Backend / Machine Learning Engineer - WIT Software
At WIT, I was responsible for turning machine learning experiments and proof-of-concept prototypes into production-ready solutions, always having performance and scalability in mind. I worked mostly in the Natural Language Processing (NLP) sub-field of Artificial Intelligence, using modern methods. My daily work encompassed all stages of research, design, implementation and testing of machine learning services (WIT Conversation Platform).
Besides research and prototyping, I was also responsible for the system architecture and the development of distributed, highly-concurrent prediction microservices. This early work has since evolved into WIT Software's agentic-AI communications platform (AINNOV8).

Technologies: Python 3.x, NumPy, Pandas, Scikit-learn, XGBoost, Keras/Tensorflow, gRPC, FastAPI, Docker, Ansible, Git.

Nov 2016 - Oct 2017

Backend Developer - Ubiwhere

At Ubiwhere, I was part of the core team responsible for developing a smart water consumption platform for the city of Porto, Portugal. This meant building a system with high-availability, multi-tenancy and concurrency support. My job consisted of implementing REST APIs and developing the core of the backend web services. Furthermore, I was also involved in the development of the frontend mobile app, used by hundreds of users and available in multiple mobile operating systems such as Android and iOS (app-aguas-do-porto).

Technologies: Django/Python, Django REST Framework, Ionic, Docker, Git.

Oct 2015 - Jul 2016

Machine Learning Research Intern - INESCTEC / CISUC

At INESC TEC/CISUC, I was involved in a research team that developed a filter system (REMINDS) that classifies public social data according to their potential relevance to a general audience, filtering out irrelevant information and relying primarily on linguistic and journalistic features. This work resulted in two peer-reviewed publications (see [1] and [2]), listed in the Publications section.

Technologies: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, PyQt5.

Oct 2013 - Mar 2014

Software Developer - Pedro Nunes Institute (IPN)

At IPN (Laboratory for Informatics and Systems), I maintained and developed new features for the information systems.

Technologies: Apache Struts/Java, Ruby on Rails/Ruby, Git.

Relevant Technical Skills

- **Programming Languages:** Proficient in Python; working knowledge of Java, C and SQL. Quick to learn new languages by focusing on transferable fundamentals.
- **AI/LLM Integration:** Building product features (information extraction, semantic/RAG search, data enrichment) on top of LLM APIs.
- **Agentic Tools & LLMs:** Claude, Gemini, Codex.
- **Web:** Frameworks: Django, Django REST Framework, FastAPI, Flask, Falcon; Markup: HTML/CSS; Protocols: REST, RPC/gRPC; Data exchange: JSON, XML.
- **Data Structures, Algorithms and Paradigms:** Solid grounding in data structures, algorithm design and optimization. Comfortable with Object-Oriented, Procedural and Distributed/Concurrent programming paradigms.
- **Version Control Systems:** Git.
- **Databases:** Relational: MySQL, PostgreSQL; NoSQL: Redis, MongoDB; Vector: Milvus, Zilliz.
- **Search & Content Infrastructure:** Algolia, Contentful.
- **Messaging Systems:** Celery with RabbitMQ and Redis as brokers.
- **Testing:** Unit, integration and end-to-end testing.
- **CI/CD Systems:** CircleCI, Jenkins.
- **Cloud Computing:** AWS (S3, EC2, ECS).
- **Infrastructure Tools:** Deployment: Docker; Configuration management: Ansible.
- **ML Tools:** scikit-learn, XGBoost, SciPy (NumPy, Matplotlib, pandas), seaborn, Keras/TensorFlow, LangChain.
- **ML & AI Concepts:** Supervised & unsupervised learning, deep learning, natural language processing (NLP).
- **Development Tools:** JetBrains Suite, Sublime Text, Unix shells.
- **Operating Systems:** Proficient in GNU/Linux and Windows.
- **Productivity/Project Management Tools:** Jira, Confluence, Slack, Trello, Redmine.

Education and training

- **University of Coimbra**
  Master Degree in Informatics Engineering - Intelligent Systems *2013 – 2016*
  - Graduated with 15/20 average
  - Dissertation titled **"Classification of Social Media Posts according to their Relevance"**
- **University of Coimbra**
  Bachelor Degree in Informatics Engineering *2010 – 2013*
  - Admission Grade: 17.55/20
  - Graduated with 16/20 average
  - Awarded to the top 3% of students (2010 – 2012)

Publications

- **Predicting the Relevance of Social Media Posts Based on Linguistic Features**
  Alexandre Pinto and Gonçalo Oliveira, H, and Alves, A., New Generation Computing, 2017 ***Springer***
- **Comparing the Performance of Different NLP Toolkits in Formal and Social Media Text**
  Alexandre Pinto and Gonçalo Oliveira, H, and Alves, A., pp 1–16, vol 51, SLATE, 2016 ***OASIcs***

Languages

- Portuguese (native), English (professional working proficiency).